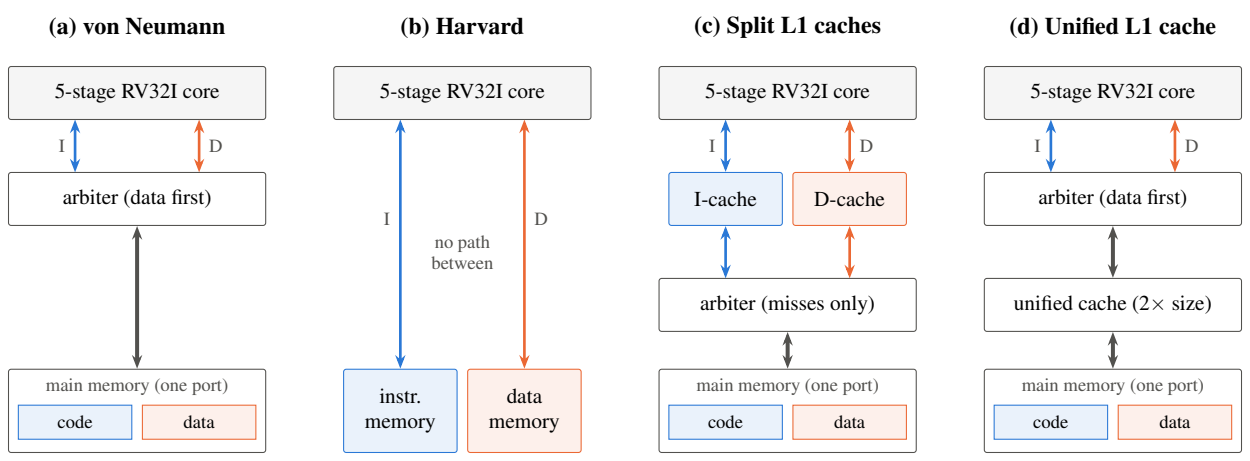
(a) von Neumann
5-stage RV32I core
I
D
arbiter (data first)
main memory (one port)
code
data
(b) Harvard
5-stage RV32I core
I
D
no path
between
instr.
memory
data
memory
(c) Split L1 caches
5-stage RV32I core
I
D
I-cache
D-cache
arbiter (misses only)
main memory (one port)
code
data
(d) Unified L1 cache
5-stage RV32I core
I
D
arbiter (data first)
unified cache (2× size)
main memory (one port)
code
data